# Early Detection of Diabetic Foot Ulcers: Deep Learning Classification using Thermograms (AlexNet Algorithm)

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Liang Wei, Chua
*Faculty of Electrical Engineering*
*Universiti Teknologi Malaysia*
Johor, Malaysia
liangwei1108@gmail.com

Nurul Ashikin Abdul-Kadir
*Faculty of Electrical Engineering*
*Universiti Teknologi Malaysia*
Johor, Malaysia
ashikin.kadir@utm.my

Nasrul Humaimi Mahmood
*Faculty of Electrical Engineering*
*Universiti Teknologi Malaysia*
Johor, Malaysia
nasrul@utm.my

Abdul Mutalib Embong
*Pusat Pendidikan Asas dan Lanjutan*
*Universiti Malaysia Terengganu*
Terengganu, Malaysia
mutalib_embong@umt.edu.my

Fauzan Khairi Che Harun
*Faculty of Electrical Engineering*
*Universiti Teknologi Malaysia*
Johor, Malaysia
fauzan@utm.my

Muhammad Amin Abd Wahab
*Faculty of Electrical Engineering*
*Universiti Teknologi Malaysia*
Johor, Malaysia
muhamadamin.abdwahab@utm.my

***Abstract*—This study aims to evaluate and improve the performance of AlexNet algorithm in classifying diabetic foot ulcer (DFU) through thermograms. Diabetes Mellitus (DM) is one of the chronic and common illness on the planet. The primary complication of diabetes is DFU. Application of deep learning neural network in diagnosing diabetics' feet can help in the early detection of DFU to prevent severe stage which may lead to amputation. In this paper, AlexNet deep learning neural networks was trained with the plantar thermogram dataset to classify the DFU. In addition, this paper also aims to investigate the effect of greyscale images in the image classification of thermogram on DFU by comparing the accuracy of image classification using colour thermograms and greyscale thermograms. The dataset used consists of 1866 coloured images which 890 images of plantar thermogram with negative DFU and 976 images of foot thermogram with positive DFU. The images were retrieved from Kaggle community with free and full access. In the experiment, data augmentation was performed on the dataset, which the 1866 images were converted into greyscale. As a result, the dataset images were doubled. At the end of this experiment, the improved network achieves an accuracy of 97.8% in classifying the thermograms.**

***Keywords—AlexNet, deep learning neural network, diabetic foot ulcer, diabetes mellitus, planar thermogram***

## I. Introduction

Diabetes, also called diabetes mellitus (DM), is a lifelong disease that cannot be cured permanently. It has become one of the major causes of mortality globally and has been on the rise for the last century [1]. In Malaysia, DM is one of the most prevalent chronic diseases, which affects around 7.8 million people. However, there are only 47% of these 7.8 million diabetic patients have been diagnosed [2].

Generally, DM comes in two main types, which are Type 1 diabetes and Type 2 diabetes [3], [4], [5]. The causes of DM include an abnormal insulin effect (Type 2 diabetes), a disturbed insulin production at the pancreas (Type 1 diabetes) and the combinational of abnormal in insulin itself and insulin secretion [6], [7]. Currently, DM can only be treated and controlled through diet, exercise, medication and lifestyle changes. With proper care and treatment, DM can go into remission at the early stage. Remission cannot be considered as cured since there is always possible for the blood sugar level to exceed the normal range and go into the diabetes range.

Diabetic foot ulcer (DFU) is one of the most prevalent complications for those with diabetes with poorly controlled. DFUs are foot injuries with chronic wound that develop in 30% of diabetic patients [8]. More admissions are result from DFUs compared to any other diabetes complication [9]. It is estimated that about 422 million people worldwide are affected by diabetes [10].

From the estimation, we can say that there will be around 127 million of people globally that will develop DFU in their lifetime. DFUs are extremely dangerous for the life of diabetics [11]. This is because diabetic ulcers are caused by a combination of poor blood circulation to deliver and provide oxygen to tissues, infection due to the open sore that expose to the ambient air or surface and primary issue of nerve damage from high blood sugar levels. Without sufficient supply of blood and oxygen to the skin tissues, the body struggles to heal its skin wounds. Due to the delay for the body to heal its wound, the skin tissues surrounding the wound will die and this can lead to necrosis, gangrene or a progression into more severe stages. In addition, foot ulcers are painful and burdened especially during the treatment and frequent clinic visits, hospitalization and replacements of wound dressing [2]. Thus, early detection of DFU is critical for aiding in diagnosing and nursing process.

Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. The examples of the processes include learning (the ability to improve performance based on experience), reasoning (the ability to draw conclusions and make decisions based on logical rules), and self-correction [12]. To add, AI systems can be trained to perform a variety of tasks, including speech recognition, object recognition in images, prediction, and game play.

Deep learning is a subfield of machine learning that is based on multiple-layer artificial neural networks (ANNs), also known as deep neural networks (DNNs). These networks are designed to learn representations from input data automatically and adaptively, typically in an unsupervised or semi-supervised manner [13]. Deep learning is used in many applications, including image recognition, natural language processing, and self-driving cars.

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In deep learning, a pre-trained network is a neural network that has been trained on a large dataset for different tasks [14]. It can be used as a starting point for training a new network for a different task. The main advantage of using pre-trained networks is that pre-trained networks have learned the useful features from the other data which can be fine-tuned to a new task. AlexNet is an example of pre-trained deep learning networks for image classification and a Convolutional Neural Network (CNN). The other examples of CNN and pre-trained deep learning networks for image classification include ResNet, VGGNet, DenseNet and MobileNet.

Some previous studies had worked on DFU classification using AI approaches such as ESNet, AlexNet and self-developed algorithm [19]-[22]. However, some of them only concentrated on the physical appearance of foot skin, while others assessed their results with a limited number of images in the testing set and a few trained their models on small datasets.

## II. METHODOLOGY

### A. Dataset

The simulation was conducted using 3732 images of foot thermograms from the Kaggle community with free access [15]. Foot thermograms are also known as plantar thermograms. The plantar thermogram is a medical image that captured using infrared technology. The temperature variations in the feet can be captured in order to observe the heat patterns to detect abnormal blood flow, nerve damage, peripheral artery disease or diabetes. In this experiment, the dataset used is originated from the plantar thermogram database, consists of 334 thermograms, 90 of which are normal foot thermograms (Class 1) and 244 of which are DFU thermograms (Class 2) [16]. The images are then undergone image augmentation through sharpening and rotation.

The dataset is divided in to three sets in order to develop and evaluate a model. The training set is used to train the model while the validation set is used to evaluate the model during training. The test set is used in evaluating the final performance of the model after the validation processes are completed. The division of the data into these three sets is critical to ensuring that the model does not overfit to the training data and can generalize well to new, previously unseen data. The common ratio used is 60-20-20 or 70-15-15 for training, validating and test set respectively. In the simulation, the dataset is divided into 60% training, 20% validation and 20% testing. The distribution of thermograms is shown in Table I.

TABLE I. DISTRIBUTION OF TRAINING, VALIDATION, AND TESTING DATA

| Sample class label | Training data | Validation data | Testing data |
|---|---|---|---|
| 1 | 600 | 200 | 200 |
| 2 | 600 | 200 | 200 |
| **Total** | **1200** | **400** | **400** |

### B. Image Preprocessing

In an effort to train the AlexNet neural network, the input images have to be resized to $227 \times 227 \times 3$. Therefore, the coding to resize the thermograms had been carried out in the project. Moreover, the thermograms had been convert into greyscale images in order to simulate and identify the effect of the greyscale thermograms to the image classification on DFU. The coding to convert thermograms into greyscale images was also implemented for this project. An example of colored foot thermogram with negative DFU and its greyscale image are shown in Fig. 1. The pre-processed images will then be loaded to the neural network for further analysis.

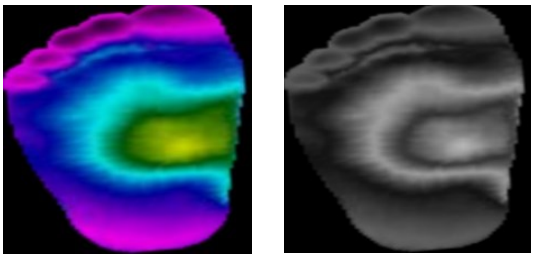

Fig. 1. Example of foot thermograms with negative DFU

### C. AlexNet Deep Learning Neural Network

AlexNet is a convolutional neural network (CNN) architecture that was introduced by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton in 2012. It was the first CNN architecture that achieved state-of-the-art performance on the ImageNet dataset, a large-scale dataset of images with 1000 classes. The architecture was designed to be deeper and wider than previous CNN architectures [17]. AlexNet comprises eight layers, including three fully connected layers and five convolutional layers. The first and second convolutional layers have a stride of 4, which reduces the spatial resolution of the feature maps by 75%, thus reducing the number of parameters. The third, fourth and fifth convolutional layers are connected by max pooling layers which reduce the spatial resolution by 2. The last three layers are fully connected layers. To add, the input image size for AlexNet should be $227 \times 227 \times 3$.

### D. Network training and hyperparameters tuning

MATLAB was used to execute and carry out the simulation tasks. Some critical hyperparameters, including epoch number, mini-batch size, and initial learning rate were manually fine-tuned. Proper selection of these values would result in improved extraction of hidden data representations. A total of 55 different combinations of these parameters were investigated in our experiment simulations. The average training time for the models was between 15 and 60 minutes. At the end of the studies, the optimum combination is identified based on these training performances. The flowchart of this work was shown in Fig. 2.

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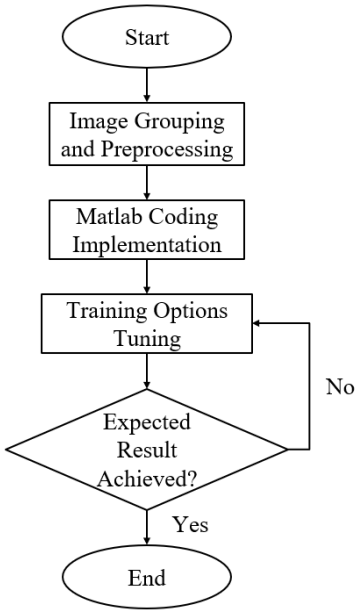


Fig. 2. Workflow of the experiment

### E. Performance metrics

The models' performance metrics were measured using classification accuracy in (1). The accuracy describes how close a prediction is to the actual value.

$$Accuracy = \frac{\sum_{i=1}^{T} \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i}}{T} \quad (1)$$

where $T$ is the total number of class labels ($T$ = 2). A true positive ($TP_i$) is a concentration class that is correctly predicted to be a specific class label, $i$. A false positive ($FP$) is a concentration class that is incorrectly predicted to be a class member, a false negative ($FN$) is a concentration class member that is misclassified as another concentration class member, and a true negative ($TN$) is a concentration class member that is correctly classified as another concentration class member.

### F. K-fold Cross-Validation

K-fold cross-validation is a machine learning approach used to evaluate the performance of a predictive model. It aids in predicting how accurately a model will perform on untested data. In the experiment, the original dataset is divided into 5 equal-sized subsets with 200 DFU positive and 200 DFU negative thermograms each. The model is trained and evaluated 10 times, where each time it is trained and tested with different subset combinations. The main advantage of k-fold cross-validation is that it provides a more robust estimation of model performance, as it uses multiple iterations and assures that all data points are considered for training and testing.

## III. RESULTS AND DISCUSSIONS

In this study, the epoch number and batch size were tuned manually to train the model. The model was run for three times for each set of same epoch number and batch size value to obtain its average performance accuracy. The mean accuracy and the uncertainty that obtained from different values of parameters are shown in Table II.

TABLE II. THE CONSIDERED HYPERPARAMETERS

| Mean ± Uncertainty (%) | | | |
|---|---|---|---|
| **Epoch number** | ***Mini Batch Size*** | | |
| | ***8*** | ***16*** | ***32*** |
| 2 | 82.50 ± 5.40 | 85.83 ± 4.25 | 87.50 ± 10.61 |
| 3 | 92.17 ± 1.64 | 94.33 ± 1.55 | 96.17 ± 1.18 |
| 4 | 92.67 ± 0.62 | 95.23 ± 0.71 | 96.27 ± 1.64 |
| 5 | 91.17 ± 5.22 | 95.30 ± 1.49 | 99.47 ± 0.47 |
| 6 | 92.93 ± 2.13 | 95.47 ± 0.90 | 97.57 ± 0.67 |

The highest classification accuracy of the model is given by 99.47 ± 0.47% which highlighted in green in Table II. Besides, the highest accuracy that obtained from the simulation also has the lowest uncertainty or standard deviation (SD). This project applies an epoch number range from 2 to 6 to examine the impact of a small number of epochs on the classification performance of a deep learning model. It is observed that even with a small value of epoch number, the deep learning model able to achieve an accuracy of 90% and above during training.

In the experiment, the trained models with the classification accuracy over 95% are accepted and saved while the trained models with the classification accuracy lower than 95% are rejected. Each of the selected trained model is then being tested with additional set of thermogram to verify and evaluate the performance of the network. The additional set of thermogram contains 780 foot thermograms with negative DFU and 952 foot thermograms of positive DFU which previously were not involved in the training process. From the evaluation, the highest obtained classification accuracy is 97.8%.

The trained model that achieves the highest obtained classification accuracy is the model that achieved the highest classification accuracy during the tuning phase. The confusion matrix in Fig. 3 shows the result of the trained model in classifying the DFU through the plantar thermograms.

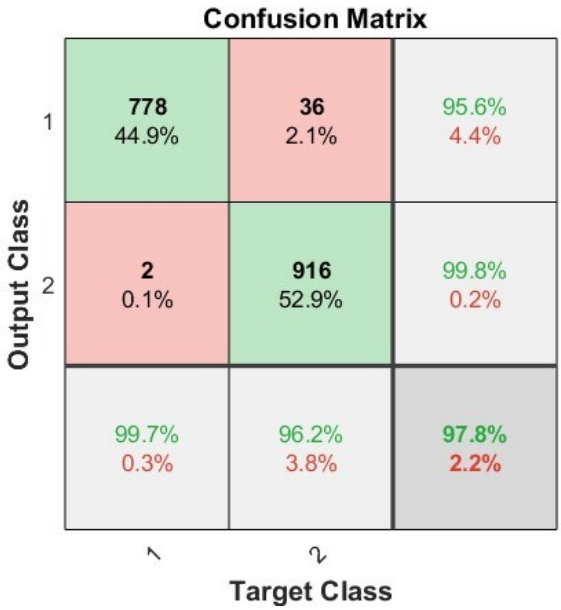


Fig. 3. Confusion matrix for the classification performance of the trained model

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TABLE III. PERFORMANCE COMPARISON WITH RECENT RELATED STUDIES

| Related study | Dataset | Colour/ Greyscale | Deep Learning Model | Accuracy (%) |
|---|---|---|---|---|
| Munadi et al. [19] | Thermograms | Colour | ESNet with Decision Fusion (DF) | 100.00 |
| Prabhu & Verma [20] | Thermograms | Colour | Self-proposed approach | 97.90 |
| Cruz-Vega et al. [21] | Thermograms | Greyscale | AlexNet | 81.18 |
| Alzubaidi et al. [22] | Foot skin images | Colour | AlexNet | 89.10 |
| This work | Thermograms | Both | AlexNet | 97.80 |

Table III presents the comparison of recent related studies on the performance of deep learning model in classifying DFU. There are very few studies about the performance of AlexNet algorithm in the field of DFU diagnosis. When comparing the performance of AlexNet architecture in the recent study with other deep learning neural networks used in the paper, it was observed that the AlexNet model exhibited relatively lower DFU classification accuracy, regardless of whether foot thermograms or normal foot skin images were used [21], [22].

Previous studies have demonstrated successful results with their approach, achieving an accuracy rate over 95% in the classification of DFU using plantar thermograms [19], [20]. However, the number of images that used to evaluate the performance of the trained model is considered insufficient. Moreover, the papers rarely discuss about the effect of greyscale and non-greyscale images in the classification of DFU. In this paper, the colored thermograms were converted into greyscale using Matlab coding through the “rgb2gray” function. The colored thermograms and greyscale thermograms were then being trained and tested seperately in order to examine their effect to the classification accuracy of DFU.

In AI training, the number of epochs, size of mini batch and initial learning rate are examples of significant parameters that affect the deep learning neural network notably [18]. In this paper, a small experiment on the comparison of the effect of different initial learning rate to the classification accuracy was done. It was found that higher learning rate mostly will reduce the classification accuracy while low learning rate requires higher training epoch number to guarantee changes to the weights of individual nodes [18]. Besides, the trained model was evaluated additionally using a different testing set which consist of 1732 images. This step provided a better guideline to verify how well the model will perform on the unseen data compared to the previous studies.

Apart from that, this study also works on investigating the effect of greyscale images in the classification of DFU. The trained model is tested with two different sets of thermograms which one of them is the colored thermograms and another is the same set of thermograms in greyscale. The classification accuracy for colored thermograms and greyscale thermograms are 98.2% and 97.5% respectively. The result shows only a slight difference in terms of classification accuracy. This indicates that the greyscale images will not significantly affect the performance of DFU classification.

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|---|---|---|---|---|
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| Cruz-Vega et al. [21] | Thermograms | Greyscale | AlexNet | 81.18 |
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| This work | Thermograms | Both | AlexNet | 97.80 |

Therefore, it is proven that the plantar thermograms can be saved as greyscale document. This can be particularly beneficial when dealing with large datasets or in resource-constrained environments, allowing for more efficient storage and retrieval of medical images. Furthermore, greyscale thermograms are still effective in capturing the necessary information for DFU classification, making them a practical and viable option for medical imaging purposes.

After achieving the enhanced result, the same dataset was split into 5 equal-sized subsets and trained 10 times with different subset combinations using the same hyperparameters that produced the best-performing trained model. In the ten separate training sessions, outcomes that achieved classification accuracy greater than 95% were saved and evaluated using the same testing set that was utilised in the previous evaluation step. However, the classification accuracy after K-fold Cross-Validation did not exceed the highest-achieved percentage of 97.8%.

## IV. CONCLUSIONS

This study used the AlexNet model to classify diabetic foot ulcer thermograms with the goal of improving classification accuracy to ease the process of diagnosing the DFU. The proposed classification model produces considerably good classification accuracy of 97.8% using a limited dataset. This demonstrates that manipulating hyperparameters also significantly impacts the classification accuracy of DFU. However, there are some limitations in this experiment, including the limited number of plantar thermograms and insufficient network layers to extract crucial local patterns of features from the input data, which could have potentially compromised overall performance of the model. Future works involve the modification of the deep learning architecture to add more convolutional layers in extracting local patterns of features that required for classification and an increased number of thermograms for the dataset to train and test the network. In short, the accuracy of classification on DFU using AlexNet model is improved through the manipulation of hyperparameters, and the result would help to ease the diagnosis process of DFU.

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